L08 ELECTRIC LIGHT QUALITY | O (MAX : 3 PT)

Intent : Enhance visual comfort and minimize flicker for electric light.

Summary : This WELL feature requires projects to take into account characteristics of electric light used in the space, such as color rendering and flicker.

Issue : Humans have evolved to depend on the sun as the main and ideal source of light. Humans are tuned to the color rendering provided by daylight and recognize colors in association with daylight.[1] Color can impact peoples' cognition and behavior.[2] Using electric light with high color rendering can improve people's perception of a space, and low color rendering can impact the ability to differentiate between objects and perceive the surroundings accurately. Electric lighting used indoors also has low frequencies of flicker that are not present in daylight. Flicker has been associated with eye strain, headaches, migraines and epileptic seizures.[3–6] In 2016, migraines accounted for 16 million disability adjusted life years (DALYs) in men and 30 million DALYs inwomen.[7]

Solutions : Identifying and utilizing lighting fixtures that emit a high quality of light and do not display signs of flicker contributes to a comfortable and healthy space. Light fixtures with higher color rendering emit light that shows colors realistically. The CIE General Color Rendering Index ("CRI") and measures from IES TM-30 are commonly used to communicate the color rendering properties of a light source.

PART 1 ENHANCE COLOR RENDERING QUALITY (MAX : 1 PT)

For All Spaces except Circulation Areas:

All luminaires in occupiable spaces (except decorative fixtures, emergency lights and other lighting for signage) meet at least one of the following color rendering requirements. If tunable white lighting is used, the requirements are met at 1000 K intervals from the lower end (with a minimum of 2700 K) to the higher end (with a maximum of 5000 K):

a. CRI (R_a) $\geq$ 90.

b. CRI (R_a) $\geq$ 80 and R9 (R_9) $\geq$ 50.

c. IES TM-30 P1 (i.e., $R_f \geq 78$, IES $R_g \geq 95$, -1% $\leq$ IES $R_{cs,h1} \leq$ 15%).[10]

For Circulation Areas:

All luminaires (except decorative fixtures, emergency lights and other special-purpose lighting) meet at least one of the following color rendering requirements:

a. CRI (R_a) $\geq$ 80.

b. P2 (i.e., IES $R_f \geq 75$, IES $R_g \geq 92$, -7% $\leq$ IES $R_{cs,h1} \leq$ 19%).

PART 2 MANAGE FLICKER (MAX : 2 PT)

For All Spaces:

All luminaires (except decorative lights, emergency lights and other lighting for signage) and their controls within occupiable spaces meet at least one of the following requirements:

a. Classified as “reduced flicker operation” per California Title 24, when tested according to the requirements in Joint Appendix JA-10.[8]

b. Recommended practices 1, 2 or 3 as defined by IEEE standard 1789-2015 LED.[9]

c. Pst LM $\leq$ 1.0 and SVM $\leq$ 0.6.

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